# Octagonal Gaming Table Turn Counter — Design Document

> **This is the advanced / permanent-install reference.** It documents the mains-powered version: a 5 V PSU on the pool-table frame, an AC switch/inlet, a slab↔frame DC disconnect, an optional level shifter, and power injection sized for full-brightness all-on. **For the current default build — USB-powered, everything on the lid, no mains wiring — start with design_doc_simple.md.** The simple build is a strict subset of this one; everything here is an additive upgrade.

| | |
|---|---|
| **Project** | LED rim turn counter for an octagonal gaming table |
| **Geometry** | 8 sides × 20" each = 160" (~4 m) perimeter |
| **Player count** | Variable, 2–8 (configurable in firmware setup mode) |
| **Input method** | Piezo sensor per side (slap-to-advance) |
| **Brain** | ESP32-S3 (with optional OTA firmware updates over Wi-Fi) |
| **Target** | Reliable, satisfying turn-passing mechanic |

> **Companion files**: - `dry_run.md` — Phase –1 bench & skills shakedown on penny parts; do this first - `turn_counter_wiring.svg` — full schematic, print and keep at the bench - `doc-src/breadboard_layout.svg` — hole-by-hole breadboard placement for the **Phase 2 bench prototype** (it includes the level shifter for learning; the Phase 0 Tap Light and the default main build both drive direct at 3.3 V — the shifter is optional, see §3.3) - `turn_counter.ino` — main project firmware - `tap_light.ino` — Phase 0 starter project firmware
>
> Other diagrams (top view, edge cross-section, slab/frame architecture) are embedded inline in the relevant sections of this document.

## Summary

A perimeter LED rim around an octagonal gaming table shows whose turn it is — one section glows in the active player's color while the rest stays dark. The active player taps their section to pass turn; the lit zone advances clockwise to the next player. Player count is configurable from 2 to 8, and a two-handed slap on opposite sides toggles the whole thing on or off.

The build is a thin octagonal slab that sits on top of an existing bumper pool table. The slab is removable — a single DC connector pops apart so it can lift away, leaving the PSU and AC mains permanently mounted to the pool table frame underneath. *(For a simpler, fully-portable build with no frame wiring at all, a USB powerbank on the slab can replace the PSU entirely — see §3.5.)*

This document walks through the build end-to-end: tools and a Phase 0 starter project for first-time soldering, bill of materials, electrical and mechanical design, step-by-step assembly, firmware reference, and troubleshooting.

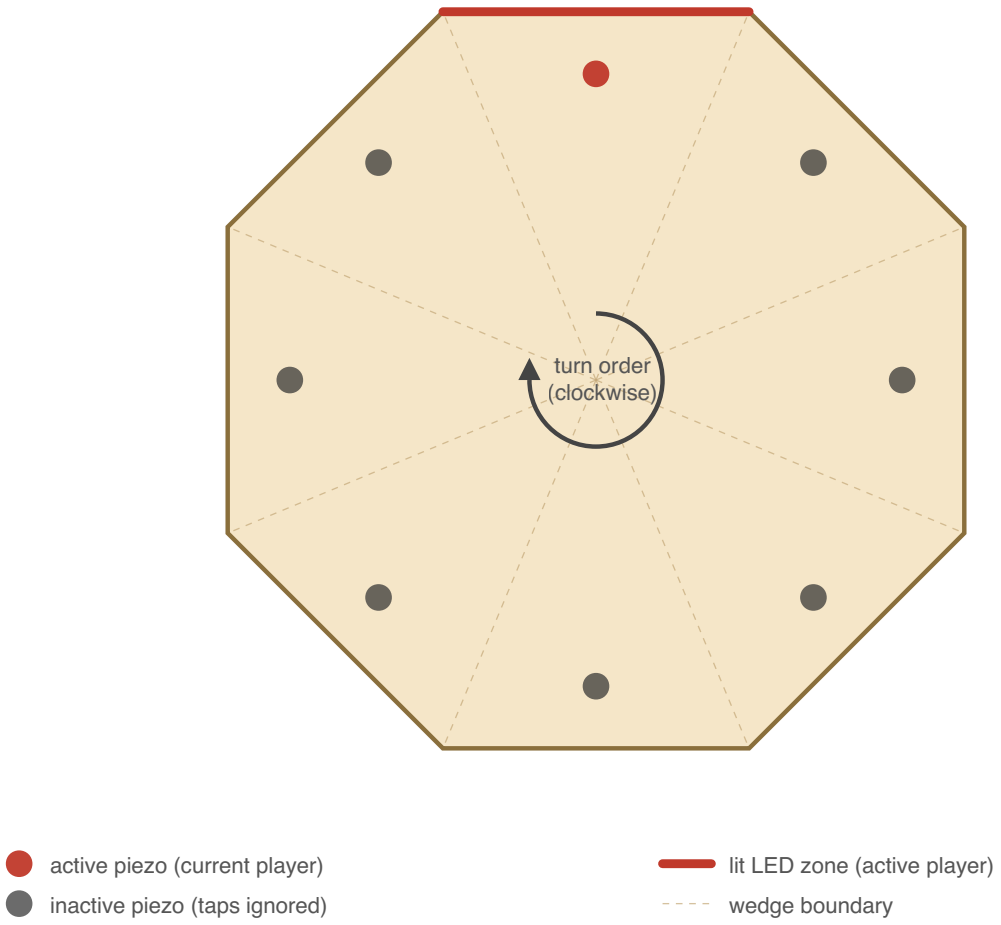



*Figure 0.1 — top-down view of the finished table. The lit edge indicates the current player's section; tapping that section advances the turn clockwise to the next player.*

# 0. Before You Start: Tools, Skills, and a Practice Project

This is your first electronics project. Don't skip this section. Going straight to the main build means doing 40+ solder joints on parts you can't easily replace, while learning to solder, on a project that won't work if any one of them is bad. That's a bad time.

Phase 0 has three goals:

1. Get you a real bench setup with the tools you'll need.
2. Teach you the techniques (soldering, multimeter use, ESP32 flashing) on a small, forgiving project.
3. Leave you with a finished, working **tap-activated desk light** that uses the same parts and patterns as the main build, just at 1/8 scale.

Plan on **4–6 hours** for Phase 0 spread over a couple of evenings. First-time soldering is slow, and that's fine.

> **Even before Phase 0**, there's a ~2–3 hour **Phase –1 dry run** in `dry_run.md`. It shakes down your bench, multimeter, soldering, and (optionally) the ESP32 toolchain on penny parts from your ELEGOO kit — no strip, no PSU, no expensive components on the bench. If this is genuinely your first time holding an iron, do it first: it means any problem on the Tap Light is a wiring issue, not a "does my equipment even work" mystery.

## 0.1 Tools You'll Need

**Essential (~$80–120 total)**:

| Tool | Recommendation | Notes |
|---|---|---|
| Soldering iron with adjustable temp | Pinecil V2 (~$30) or Hakko FX-888D (~$110) | Skip the $15 fixed-temp pencil irons. Variable temp is the difference between fun and frustration |
| Solder | 60/40 or 63/37 leaded rosin core, 0.6 mm or 0.8 mm | Leaded is much easier to learn on (63/37 is eutectic — snaps solid instantly, even fewer cold joints). Ventilate and wash hands. Lead-free is harder and not necessary at this scale |
| Side cutters / flush cutters | Hakko CHP-170 (~$8) | For trimming component leads |
| Wire strippers | Any self-adjusting strippers (~$15) | Not the manual ones with a sliding stop |
| Multimeter | Any cheap one (~$20) | You only need continuity and DC voltage. AstroAI on Amazon is fine |

**Strongly recommended (~$30 total)**:

| Tool | Recommendation | Notes |
|---|---|---|
| Helping hands or PCB vise | Anything with alligator clips and a base | Genuinely transformative. Soldering with no third hand is miserable |
| Flux pen or paste | Kester 951 or MG Chemicals | Makes solder flow properly. You need this |
| Heat-shrink assortment | $10 mixed pack | Insulating splices and joints |
| Lighter or hot air gun | Any Bic | For heat-shrink. Hair dryer doesn't get hot enough |
| Solder wick / desolder braid | $5 | For fixing mistakes (you will make some) |
| Bench mat | Silicone, ~$15 | Heat-resistant, prevents you from burning your desk |

**Nice to have, but skip for now**: fume extractor, hot air rework station, magnifier, "third hand" with magnifier.

## 0.2 Soldering Primer

The skills are mostly muscle memory. Watch a video before you start — these two are good:

- "How to Solder" by EEVblog (12 min — fundamentals)
- "Common Soldering Mistakes" by Branchus Creations (8 min — what to avoid)

The condensed version that text can convey:

1. **Set the iron to ~330°C / 625°F** for leaded solder. Higher for lead-free, but you're using leaded.
2. **Tin the tip** when hot — touch a bit of solder to the tip, wipe on a damp sponge or brass coil. The tip should look shiny silver. A black, dull tip transfers heat poorly.
3. **Heat the joint, not the solder.** Touch the iron tip to both pieces being joined. Wait ~2 seconds. Then feed solder into the joint *from the opposite side of the iron*. The solder should flow toward the heat and wet both surfaces.
4. **Pull the solder away first, then the iron.** Don't move the joint while it cools (~1–2 seconds).
5. **Good joint**: shiny, smooth, slightly concave, wets both surfaces. **Bad joint**: dull, blobby, ball-shaped, only stuck to one side ("cold joint").
6. **If a joint is bad**, add a touch of flux, reheat, and either reflow or remove with wick and redo.

**Warmup exercise before touching real parts**: strip 4–5 pieces of scrap wire (any gauge), twist pairs together, and solder them. You'll do ~10 joints. By the last one, they should look noticeably better than the first. Don't proceed to real components until you can produce consistent shiny joints.

## 0.3 Multimeter Primer

You need exactly two functions for this project:

**Continuity** (often a 🎵 or diode symbol). Probes touched together = beep. Use it to:

- Verify a wire goes from where you think it does to where you think it does
- Check that solder joints actually connected
- Confirm a fuse is intact

**DC voltage** (V with a straight line, often "V⎓" or "VDC"). Set the range to 20V or use auto-range. Use it to:

- Verify a 5V power supply is putting out 5V (and not 12V because you grabbed the wrong PSU)
- Check polarity (red probe to +, black to GND, expect a positive number)

**Critical habit**: every time you connect power to something new, multimeter the polarity at the destination side first. Reversed 5V on a WS2812B strip kills LEDs instantly and silently.

## 0.4 Arduino IDE Setup

Before any soldering, get the dev environment working:

1. **Install Arduino IDE 2.x** from arduino.cc.
2. **Add ESP32 board support**: File → Preferences → Additional Board Manager URLs → add `https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json`. Then Tools → Board → Boards Manager → search "esp32" → install "esp32 by Espressif Systems". (Same package supports the ESP32-S3.)
3. **Install libraries**: Tools → Manage Libraries → install **FastLED** (by Daniel Garcia). For the main project later, you'll also install **ArduinoOTA** (already bundled with the ESP32 core, no separate install).
4. **Plug in your ESP32-S3 dev board** via USB-C. Most DevKitC-1-style boards have **two** USB ports and either one flashes the chip: the port labeled **USB** is the S3's native USB (enumerates as a USB CDC device, `/dev/cu.usbmodem*` on macOS — no driver to install); the port labeled **UART**/**COM** goes through a CP2102 bridge ( `/dev/cu.usbserial-*` — modern macOS and Windows ship the driver). Serial output follows the **USB CDC On Boot** setting: Enabled → native port, Disabled → UART port.
5. **Test upload**: File → Examples → 01.Basics → Blink. Tools → Board → ESP32 Arduino → **"ESP32S3 Dev Module"**. Tools → **USB CDC On Boot: Enabled** (so Serial reaches the IDE over the native USB port). Tools → Port → select the new port that appeared when you plugged in the board. Click upload. If your board has an onboard RGB LED, the Blink example won't drive it — try `Examples → ESP32 → ChipID` to confirm the chip responds.

If upload fails: hold the BOOT button on the board, tap RESET, release BOOT, then upload. Drop the upload speed to 115200 in Tools → Upload Speed if it still fails.

> **Before starting the Tap Light**: if you haven't done the **Phase –1 dry run** ( `dry_run.md` ), do it now. It confirms your iron, meter, soldering, and the ESP32 toolchain all work — on throwaway parts — so the Tap Light is the first time those skills meet real components, not the first time you find out a tool is broken.

## 0.5 The Starter Project: Tap Light

A small WS2812B accent light with a single piezo sensor. Tap it to cycle through 6 modes (warm orange, blue, green, amber, rainbow, off). Power it from the ESP32's USB port — no separate PSU, no mains wiring, no level shifter to start.

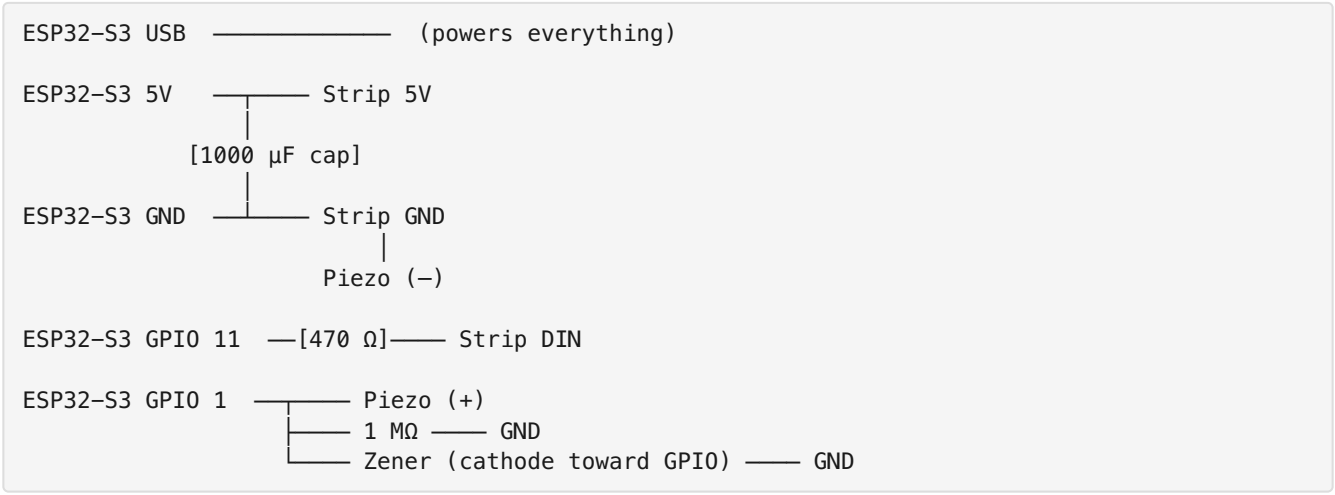
This is essentially **one zone of the main project**. Every skill you exercise here transfers directly.

### Parts (all from the main BOM)

| Qty | Part |
|---|---|
| 1 | ESP32-S3 dev board |
| 30 LEDs | WS2812B strip (cut from the 5 m roll — you have plenty) |
| 1 | Piezo disc |
| 1 | 1 MΩ resistor |
| 1 | 3.3 V Zener diode |
| 1 | 470 Ω resistor |
| 1 | 1000 $\mu$F capacitor (optional at this scale, but install it for practice) |
| ~ | Hookup wire, heat-shrink |
| 1 | Breadboard (~$5, get one anyway, useful for life) |
| 1 | USB cable for the ESP32 |

Skip the level shifter for the starter. WS2812B often works fine on 3.3 V data directly — the main build relies on this too (direct 3.3 V drive is the default; the level shifter is an optional add-back if the first pixel ever misbehaves — see §3.3).

### Wiring




This is identical to the main wiring diagram, just with one piezo channel and no level shifter.

### Build Steps

#### Session 1: Solder practice + LED strip prep (1–2 hours)

- ☐ Set up your bench: iron heating, solder, helping hands, multimeter, scrap wire
- ☐ Warmup: 5 wire-to-wire solder joints on scrap. Inspect each. Don't proceed until they look good
- ☐ Cut a 30-LED segment from the strip (between LED pads, follow the cut lines printed on the strip)

☐ Tin the three pads at one end of the segment (5V, DIN, GND). Add solder until each pad has a small dome

☐ Strip ~4 mm of insulation off three short hookup wires (red for 5V, black for GND, any other color for data)

☐ Tin the wire ends

☐ Solder each wire to its pad. Touch iron to pad, melt the dome, slide wire in, hold steady, remove iron. ~2 seconds per joint

☐ Continuity test: probe each wire end against the matching pad on the strip. Should beep

☐ Heat-shrink each joint individually for strain relief

**Session 2: Build the input network and breadboard (30–60 min)**

☐ Solder leads to the piezo: red to (+), black to (–). Be **fast** — under 2 seconds per pad. Piezo discs lose sensitivity when overheated

☐ On a breadboard, lay out the circuit per the wiring above

☐ Place the ESP32 across the breadboard's center channel

☐ Use jumper wires for the ESP32-to-component connections

☐ The 1 MΩ resistor sits between GPIO 1 and GND

☐ The Zener sits between GPIO 1 and GND, with the **black band (cathode) toward GPIO 1**, body toward GND. Polarity matters

☐ Don't connect power yet

**Session 3: Test and tune (1 hour)**

☐ Visual check: walk the wiring against the schematic one more time

☐ Multimeter polarity check: with USB unplugged, probe ESP32 5V to GND with continuity — should NOT beep (no shorts)

☐ Plug in USB. The onboard power LED should light (most ESP32-S3-DevKitC boards have one near the USB port). If you smell anything or the board gets warm, unplug immediately and check for shorts

☐ Open Arduino IDE, load `tap_light.ino`, upload

☐ Strip should light up in the first mode (warm orange). The onboard RGB pixel mirrors the current mode too, so tap → color cycling is testable even before the strip is wired

☐ Open Serial Monitor at 115200 baud — at boot it prints the piezo baseline (the sketch averages the resting level for ~0.5 s, then keeps tracking it slowly)

☐ Tap the piezo. Mode should advance, and a line should print to serial showing the reading, the baseline, and the delta between them

☐ If no response: tap harder, or lower `TAP_DELTA` in the code and re-upload. Detection is adaptive — a tap fires when a reading jumps `TAP_DELTA` (default 720) above the tracked baseline, so noise and drift are absorbed automatically; `TAP_DELTA` only sets how hard a hit must be

☐ Cycle through all 6 modes by tapping. Confirm the rainbow mode looks right (each LED a different color)

☐ Unplug USB, plug back in. The strip should resume on whatever mode you left it on (state persists in NVS)

**Session 4 (optional): Move from breadboard to protoboard, add an enclosure**

If you want a finished thing rather than a breadboard prototype:

- ☐ Lay out the circuit on a small protoboard (5×7 cm)
- ☐ Use female headers for the ESP32 so you don't solder it directly
- ☐ Solder the resistor and Zener through-hole; trim leads with side cutters
- ☐ Run wires to the strip and piezo via JST connectors or solder direct
- ☐ Mount in a small project box, or hot-glue everything to a piece of scrap wood

## 0.6 Skills Check

If you can honestly check all of these, you're ready for the main build.

- ☐ My solder joints look shiny and properly wetted (not blobby or dull)
- ☐ I can solder a wire to an LED strip pad without lifting the pad
- ☐ I soldered a piezo without killing it
- ☐ I know how to test continuity with my multimeter and have done so on real circuits
- ☐ I successfully uploaded firmware to the ESP32 via USB
- ☐ I read serial output and used it to tune a parameter (the piezo tap delta)
- ☐ I understand the per-piezo input network well enough that I could draw it from memory
- ☐ My Tap Light works end-to-end and persists state across power cycles

**If any of these are no**, repeat the relevant section before moving on. The main build has 8× the joints, mains wiring, and corner soldering on a strip mounted in aluminum channel — none of which is forgiving.

# 1. The Broad Plan

The main build splits into seven phases on top of Phase 0. Do them in order.

| Phase | What happens | Why it's here |
|---|---|---|
| **–1. Dry run** | Bench + skills shakedown on penny parts (see `dry_run.md`) | Prove your tools and hands work before any expensive part is on the bench |
| **0. Starter project** | Build the Tap Light (see §0) | Skills, bench setup, confidence |
| **1. Plan & gather** | Parts list ordered, work area set up | Nothing worse than getting halfway and realizing you're missing a Zener |
| **2. Bench prototype** | ESP32 + 1 LED segment + 1 piezo on a breadboard, firmware running | Proves the full firmware works before anything is permanent |
| **3. LED rim** | Cut, route, and mount the strip into aluminum channel around the octagon | The most visible part of the build. Take your time on the corners |
| **4. Piezo mounting** | Glue 8 piezos to the underside of the slab, run leads | Cross-talk between sides is inherent to a single slab; firmware filters it, but careful placement helps |
| **5. Control box** | ESP32 + 470 Ω + 1000 $\mu$F cap on a half-size protoboard, in an enclosure (level shifter optional) | Everything terminates here. In-line strip + piezo JSTs so the whole control box unplugs |
| **6. Final assembly** | Mount PSU on the bumper pool frame, wire the Powerpole disconnect, dress wiring on the slab, mate and test | The slab/frame split (§4.6) is most of the work here |
| **7. Tune & test** | Dial in the piezo tap delta, calibrate brightness, play a real game | Real-world testing always reveals something the bench didn't |

**Estimated time**: ~2–3 hours for the Phase –1 dry run, 4–6 hours for Phase 0, plus 10–14 hours for Phases 1–7 spread over a weekend. Most of the main build is mechanical (channel cutting, mounting), not electronics.

# 2. Bill of Materials

Quantities below cover the **main project**. The starter project (Phase 0) uses a subset — see §0.5. Order everything together; the per-unit prices on extras are negligible and you'll appreciate having spares.

| Qty | Part | Specifics | Notes |
|---|---|---|---|
| 1 | ESP32-S3 dev board | ESP32-S3-DevKitC-1, N8R8 variant (8MB flash + 8MB octal PSRAM), USB-C native | Buy 2 — one for the starter, one for the main. ~$15 each. N8 and N8R2 are end-of-life at DigiKey; N8R8 is the current stocked variant. Octal PSRAM uses GPIOs 35/36/37 internally — fine for this build (the pinout in §3.1 doesn't touch them), just don't reassign anything to those pins. Get the PCB-antenna part (no `U` suffix); the `-1U-N8R8` variant has a U.FL connector and no antenna |
| 5 m | WS2812B LED strip | 60 LEDs/m, IP30, 5V | 4 m used; the extra is for the starter and mistakes |
| 4 m | Aluminum LED channel | With frosted/diffuser cover | Massively improves the look |
| 10 | Piezo disc | 27 mm brass, 2 leads | Bag of 10 covers main + starter + spares |
| 12 | Resistor | 1 MΩ, 1/4 W | Pulldown across each piezo. Buy a 100-pack, they're nothing |
| 12 | Zener diode | 3.3 V, 1 W (1N4728A) | ADC overvoltage clamp |
| 5 | Resistor | 470 Ω, 1/4 W | Series resistor on data line |
| 3 | Capacitor | 1000 $\mu$F, 10 V or higher, electrolytic | Across 5V/GND at strip start |
| 2 | Level shifter *(optional)* | 74AHCT125 (DIP-14) | 3.3 V → 5 V data line — optional robustness add-back (§3.3); default build drives direct |
| 1 | Power supply | Mean Well LRS-100-5 (5 V, 18 A) | Don't cheap out here. LRS-50-5 (5 V, 10 A) is the original spec and also fine if you can find it; the -100 was substituted when the -50 went out of stock at DigiKey |
| 1 | Power switch | Panel-mount rocker rated for your mains (≥125 V AC covers US 120 V; 250 V AC if outside the US), 6 A+ | Or use a switched IEC inlet (safer) |
| ~ | Hookup wire | 22 AWG stranded for signal, 18 AWG for power | A few colors helps |

| Qty | Part | Specifics | Notes |
|---|---|---|---|
| 12 | JST-XH 2-pin connector pairs | Inline disconnect in each piezo lead, off-board (8 used + spares) | Any disc — or the whole control box — unplugs without desoldering |
| 2 | JST-XH 3-pin connector pair | For the LED strip feed (5V + data + ground) | Detachable rim |
| 1 pk | WAGO 221-415 lever connectors | 5-conductor, 25-pack | Branch nodes for the slab DC rail (control-box feed + 3 strip injection points). Replaces the screw-terminal blocks of earlier drafts |
| 1 | Project box | ~112×62×31 mm (e.g. Hammond 1591B) | Holds the 81 mm half-size control board — see §4.5 |
| 2 | Protoboard | 5×7 cm (Phase 0 starter) + half-size Perma-Proto 81×46 mm (main) | 30 columns suffice because the piezo networks live at the discs (§4.3) and the pigtails land directly on the board — no board-mounted piezo headers (their service JSTs are inline, off-board). See `doc-src/protoboard_half_layout.svg` |
| 2 | Female header strip, 1×22 | 2.54 mm pitch, cut from 1×40 strips (cut through position 23 — cutting a female header sacrifices one socket) | ESP32 socket on the control board — never solder the DevKit directly |
| 1 | Inline fuse | 5 A automotive blade fuse + holder | Between PSU 5V output and Powerpole pigtail |
| 1 | Breadboard | Standard half-size | For prototyping. Useful forever |
| | | | |
| | **Removable-top installation (§4.6)** | | |
| 3 | Anderson Powerpole 30 A connector kit | Pair of red + black housings + contacts | One pair for the slab, one for the frame side it mates with, one for a bench-test pigtail (see §4.6). Contacts get soldered + heat-shrunk, not crimped (procedure in `shopping_list.md`) — no crimp tool needed |
| 6 ft | Silicone-insulated wire | 14 AWG, red and black (3 ft each) | DC run from PSU to slab |

| Qty | Part | Specifics | Notes |
|---|---|---|---|
| 1 | Soft rubber grommet | ½″ or ⅝″ panel-hole size; inside diameter sized to your DC cable (the 14 AWG silicone twisted pair is ~$^{5}/_{16}$″ / 8 mm OD) | Cable entry through slab. Hardware-store generic — *avoid* sheet-metal-panel bushings like Heyco 2092, which are designed for 3 mm panels, not 1″ wood |
| 2 | P-clips or adhesive cable mounts | Various sizes | Cable strain relief inside slab |
| 1 | Pine block or scrap | ~4×6×1" | PSU mounting platform on frame |
| ~ | #8 wood screws | 3/4" and 1.5" lengths | Mount block to frame and PSU to block |

**Total cost**: ~$320–330 for parts, plus ~$213 for tools if you own none — `shopping_list.md` has the vendor-priced, per-part accounting. Tools are a one-time investment that'll serve every future project.

**Where to source**: Strip and channel from BTF-Lighting or Adafruit. Piezos, resistors, Zeners, level shifter from any electronics supplier (DigiKey/Mouser if you want quality, Amazon if you want speed). PSU from Mean Well's authorized resellers — there are a lot of fakes on Amazon.

**AC mains wiring note**: the BOM uses a **discrete-component approach** — a separate IEC C14 inlet and panel-mount SPST switch wired in series on the Line conductor between the wall plug and the PSU. The wall plug feeds the IEC inlet's Line and Neutral pins; Line goes through the SPST switch to the PSU's L input; Neutral runs from the IEC inlet direct to the PSU's N input (no switch); Earth runs from the IEC inlet's Earth pin direct to the PSU's chassis Earth lug. **The BOM's inline 5 A blade fuse is a DC-side part** — it goes between the PSU's +5 V output and the Powerpole pigtail (§3.3), never on the AC line: automotive blade fuses are rated 32 V DC and cannot safely interrupt 120 V AC. The AC side needs no added fuse — the LRS-100-5 carries its own internal input fuse. With the switch off, Line is interrupted so no current can flow — same approach as every wall switch in your US household. **Important safety habit:** unplug from the wall before opening the enclosure. Never rely on the switch alone as service isolation — only the wall plug guarantees both Line and Neutral are disconnected. The alternatives — an integrated switched/fused IEC inlet (Schurter DG12 series, ~$46) or a DPST switch that interrupts both Line and Neutral — are nice-to-haves, not requirements. The SPST + "unplug before service" discipline is the residential-wiring standard.

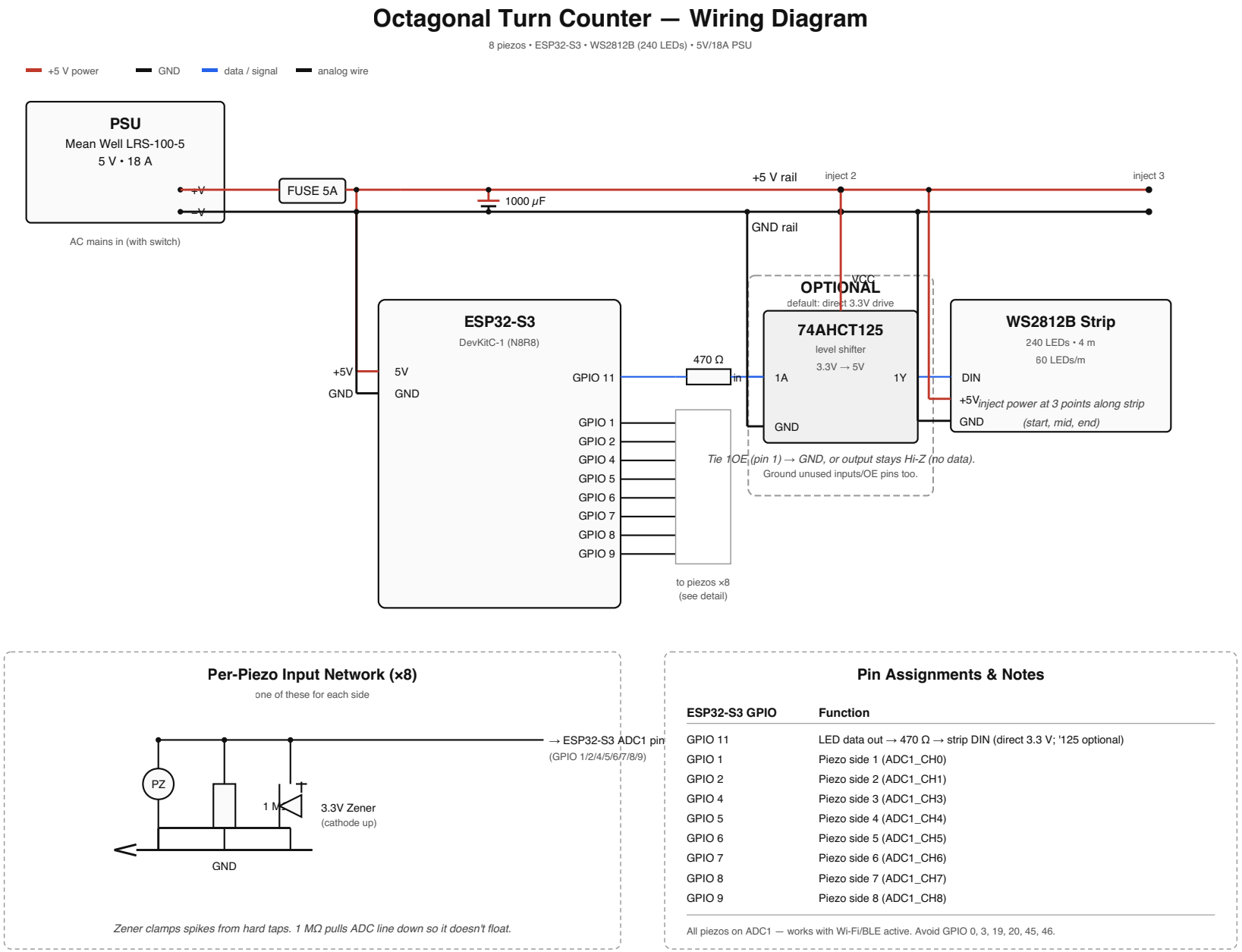



*Figure 3.1 — full wiring diagram. PSU at top-left feeds +5V/GND rails across the top. ESP32 drives 8 piezo inputs and one LED data line through a 470 Ω resistor and 74AHCT125 level shifter. Three power injection points distribute current along the strip. Per-piezo input network detail at bottom-left.*

# 3. Electrical Design

> **See `turn_counter_wiring.svg` for the full schematic.** This section summarizes; the SVG is authoritative.

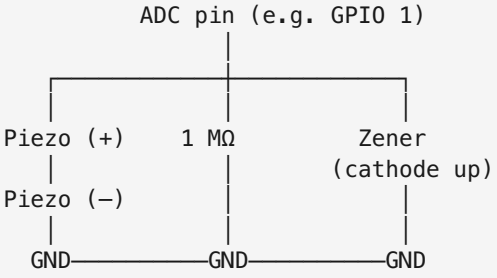
## 3.1 Pinout

| ESP32-S3 GPIO | Function | Notes |
|---|---|---|
| 11 | LED data out (→ 470 Ω → strip DIN, direct 3.3 V) | |
| 1 | Piezo side 1 ADC | ADC1_CH0 |
| 2 | Piezo side 2 ADC | ADC1_CH1 |
| 4 | Piezo side 3 ADC | ADC1_CH3 |
| 5 | Piezo side 4 ADC | ADC1_CH4 |
| 6 | Piezo side 5 ADC | ADC1_CH5 |
| 7 | Piezo side 6 ADC | ADC1_CH6 |
| 8 | Piezo side 7 ADC | ADC1_CH7 |
| 9 | Piezo side 8 ADC | ADC1_CH8 |

All 8 piezos are on **ADC1**. This matters: the ESP32-S3 radio takes over **ADC2** when Wi-Fi or BLE is active, and any `analogRead` on an ADC2 pin silently returns 0. Sticking to ADC1 (GPIOs 1–10) means tap detection keeps working with OTA, a phone web UI, or any future BLE feature.

Avoid GPIO 0, 3, 45, 46 — strapping pins; pulling them at boot changes boot mode (GPIO 0/46), flash/VDD_SPI voltage (GPIO 45), or the JTAG signal source (GPIO 3). GPIO 19 and 20 are the native USB D-/D+ lines — leave them untouched. GPIO 35/36/37 are reserved on Octal-PSRAM modules (N8R8 / N16R8) — and since the BOM now specs the N8R8 (N8/N8R2 are EOL at DigiKey), those three pins are off-limits for this build. The current pinout above doesn't use them, so this is only a constraint if you add features later.

## 3.2 Per-piezo circuit (×8)



The 1 MΩ resistor pulls the ADC line down so it doesn't float. The Zener clamps spikes — a hard slap on a piezo can produce 20+ volts momentarily and the ESP32-S3 ADC tops out at 3.3 V.

**Optional hardening**: a 10 kΩ–47 kΩ resistor in series between the piezo/clamp node and the ADC pin. The piezo's negative half-swing forward-biases the Zener at about –0.6 V, slightly past the S3's –0.3 V absolute-maximum rating. The classic knock-sensor circuit survives this in practice (microjoule pulses from a very high-impedance source), but the series resistor limits the injected current to tens of microamps and makes it unambiguously safe — and costs nothing for tap detection, since the ADC input draws no current.

## 3.3 LED strip + power

Three injection points along the strip — start, middle, and end — each tapping the +5V and GND rails on the slab. This prevents voltage droop along the 4 m run. Note that those rails on the slab come from the PSU on the bumper pool frame via the Powerpole DC disconnect; see §4.6 for the full slab/frame architecture. The schematic shows the electrical connections; the disconnect itself is a physical break in the +5V/GND wiring between PSU and slab.

**Data line — direct 3.3 V drive (default).** GPIO 11 drives the strip's DIN through a 470 Ω series resistor, placed close to DIN. WS2812B specifies a data "high" of ~0.7 × VDD = 3.5 V at a 5 V supply, so the ESP32's 3.3 V is just under spec — but only the **first pixel** is at risk of not latching (it regenerates a clean 5 V signal for all 239 downstream), and with the strip start sitting right by the control box the data run is short and reliable. This is the default build: simplest, and adding protection later is trivial.

**Optional robustness upgrade — 74AHCT125 level shifter.** If the first pixel ever glitches (a cold room, a replacement strip from a different batch, marginal timing), splice a 74AHCT125 into the data path: GPIO 11 → '125 input → '125 output → 470 Ω → DIN. Power VCC from the +5V rail and tie 1OE (and any unused channel OE pins) to GND, or the output stays Hi-Z. It bumps the 3.3 V logic to a clean 5 V swing with full margin. The board layout (`doc-src/protoboard_half_layout.svg`) marks where it drops in; everything stays accessible so it's an add-back, not a rebuild.

A 1000 $\mu$F cap across the 5V/GND rails right where the strip starts. Acts as a local energy reservoir for sudden current draw.

An inline 5 A fuse between PSU 5V output and the rest of the system. If something shorts, the fuse blows instead of the strip.

## 3.4 Power budget sanity check

- 240 LEDs × 60 mA worst case (full white) = 14.4 A
- One side (30 LEDs) lit at 50% brightness, single color = ~0.5 A — this is normal turn_counter play
- All active seats lit at once (setup blink / READY mode / ready-flash, up to 221 LEDs) = ~3–4 A peak — the firmware's `MAX_POWER_MA` cap (§3.5) holds this down on a current-limited supply
- Idle (lights off, ESP32-S3 only) = ~30–50 mA without Wi-Fi, ~80–120 mA with Wi-Fi up for OTA

An 18 A PSU (LRS-100-5) is generous — even a 10 A PSU (LRS-50-5) would be comfortable, since we'll never light more than ~30 LEDs at once during normal play. The extra headroom on the -100 means the PSU itself won't enter overcurrent protection even if the firmware ever drives full-brightness white across the full strip (~14.4 A theoretical max), so the **inline 5 A fuse is what protects downstream wiring on a short** — don't skip it.

## 3.5 Alternative power: through the board's USB

The frame-mounted PSU + Powerpole architecture (§4.6) is sized for the ~14 A a full-white strip could pull. But **turn_counter never does that** — normal play lights exactly one side (~27–29 LEDs), so the real draw is **under 1 A** (one side at brightness 128 ≈ 0.5–0.7 A of LEDs, plus ~0.25 A for the ESP32-S3 with Wi-Fi). That's small enough to power the **entire table through the ESP32-S3 dev board's own USB jack** — from a USB powerbank on the slab, a USB wall adapter, or even a laptop. No PSU, no mains work, no frame wiring, no chopped cables: one USB cable into the board, and the strip's 5 V comes off the board's `5V` pin. Everything lives on the slab.

**It's a current limit, not a voltage limit.** 5 V is nominal for both the board and the strip. What's marginal is the *amps* through the board's USB-C connector and the thin 5 V trace to the `5V` pin — good for roughly **1.5–2 A continuous**. Normal one-side play sits at ~0.85 A total, comfortably under that. (It's why bench-powering the strip off a laptop has been fine all along, and why `tap_light`'s own 700 mA cap keeps *it* safe through USB too.)

**The one stress case, handled in firmware.** Three moments light *every* active seat at once — setup (joined seats blinking), READY mode with everyone on, and the ready-flash — which at full brightness would be ~3–4 A, past the board's path. `MAX_POWER_MA` (default **1500**) drives `FastLED.setMaxPowerInVoltsAndMilliamps(5, 1500)`, so FastLED transparently dims *only those all-on frames* (~⅓ brightness) to hold the draw in bounds; one-side play never hits the cap and stays full-bright. 1500 mA of LEDs + ~250 mA ESP ≈ 1.75 A total — sized for the board's USB path.

**Wiring — the whole thing:**

- One USB cable from the source (powerbank / wall adapter / laptop) into the board. The strip's 5 V + GND tap the board's `5V` / `GND` pins; data stays GPIO 11 → 470 Ω → DIN; common ground throughout.
- Keep the 1000 $\mu$F cap at the strip start (§3.3).
- That's it — nothing crosses to the frame.

**Want full-brightness all-on?** (bright setup / READY) Then don't route that current through the board. Chop a USB cable for its 5 V + GND and **split it: strip injection point(s) on one branch, the board's `5V` pin on the other**, common ground — so the LED current bypasses the board entirely. Add the mid/end injection taps for the READY-all-on case (short jumpers from the same source along the slab). Then raise `MAX_POWER_MA` to ~2500 (a 3 A USB port), or remove it on the Mean Well PSU. This split is the *only* reason to chop a cable; the default one-cable path covers normal play fully.

**If your source is a powerbank, two caveats:**

- **Low-load auto-shutoff.** Idle/off draw drops to ~0.1–0.25 A; many banks read that as "nothing plugged in" and power down after ~15–30 s. Use a bank with an "always-on" / low-current / trickle mode.
- **Runtime is a non-issue.** A 10,000 mAh bank ≈ ~9 h of normal one-side play.

**Note:** the `tap_light` and `all_white` test firmwares *do* light the whole strip — `all_white` at 2000 mA will exceed the board's USB path, so run it from a stouter supply or lower its cap; `tap_light` at 700 mA is fine through USB.

# 4. Mechanical Design

## 4.1 LED channel layout

The aluminum LED channel mounts to the **outer edge of the slab**, running around the entire octagonal perimeter. The diffuser faces outward so the glow is visible from any seat at the table.

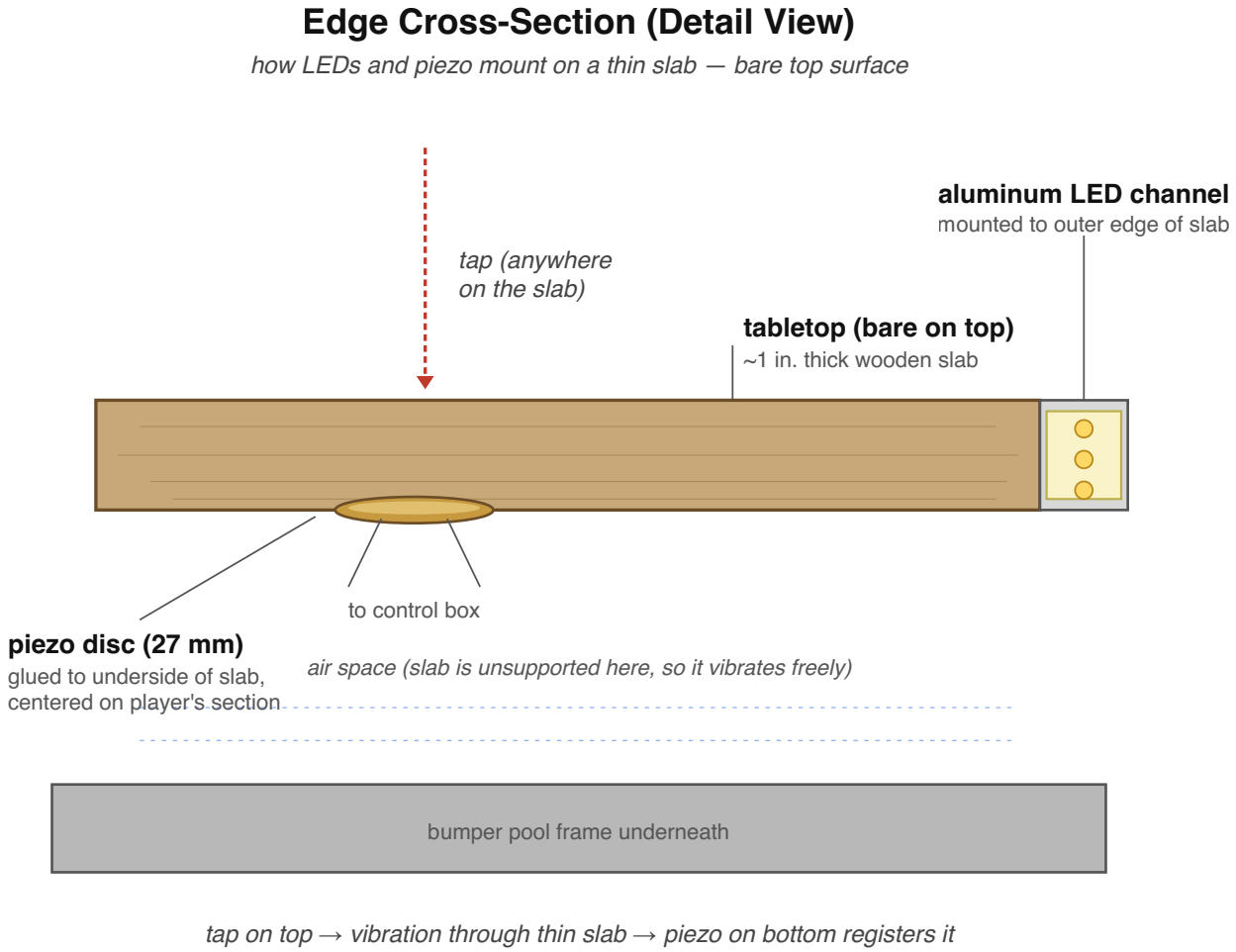



*Figure 4.1 — edge cross-section. The aluminum LED channel mounts to the outer edge of the slab; the piezo glues to the underside, centered on each player's wedge of the octagon. Air space below means the slab vibrates freely, so a tap anywhere on the bare top surface couples efficiently into the piezo underneath.*

The slab is thin (~1") and rests on the bumper pool frame with air space between, which means it vibrates freely. This is exactly what lets piezos on the underside register taps from anywhere on the slab.

Alternative: if you prefer a recessed look, route a 1/2"-wide × 1/4"-deep rabbet into the outer edge of the slab and recess the channel flush. More work, looks cleaner. Your call.

## 4.2 Corner handling

8 sides means 8 closed-loop interior corners. At each corner:

- **Cut the channel** at 22.5° miters with a hacksaw or chop saw. The angle for a regular octagon is 135° interior, so each piece gets a 22.5° cut on each end.
- **Cut the LED strip** between LED pads at the corner. Solder short jumper wires (3-conductor: 5V, GND, Data) to bridge across the corner gap. Heat-shrink the joints.
- **LED count per side**: at 60 LEDs/m, a 20" (508 mm) side fits ~30 LEDs. Cut cleanly between pads, you'll lose maybe 1–2 LEDs in the corner gaps.

**Corner caps & edge-resting protection.** The slab gets stored leaning on its side (that's how the bumper pool underneath gets played). With the channel on the outer edge, that would rest the slab's weight directly on the diffuser and the LEDs behind it — and eventually pop the channel off. The fix is corner caps standing **~5 mm proud of the diffuser face** at all 8 corners: leaned on any side, the slab lands on the two caps flanking that side and the channel never touches the floor. The caps also round off the sharp 135° wood corners and shield the corner jumper bridges — the most fragile joints in the build. Design rules

regardless of which option below you pick: **screw, don't glue** (you'll want the cap off to service the joint behind it), and give the landing surfaces a grippy rubber skin so the leaned slab grips the floor instead of skating.

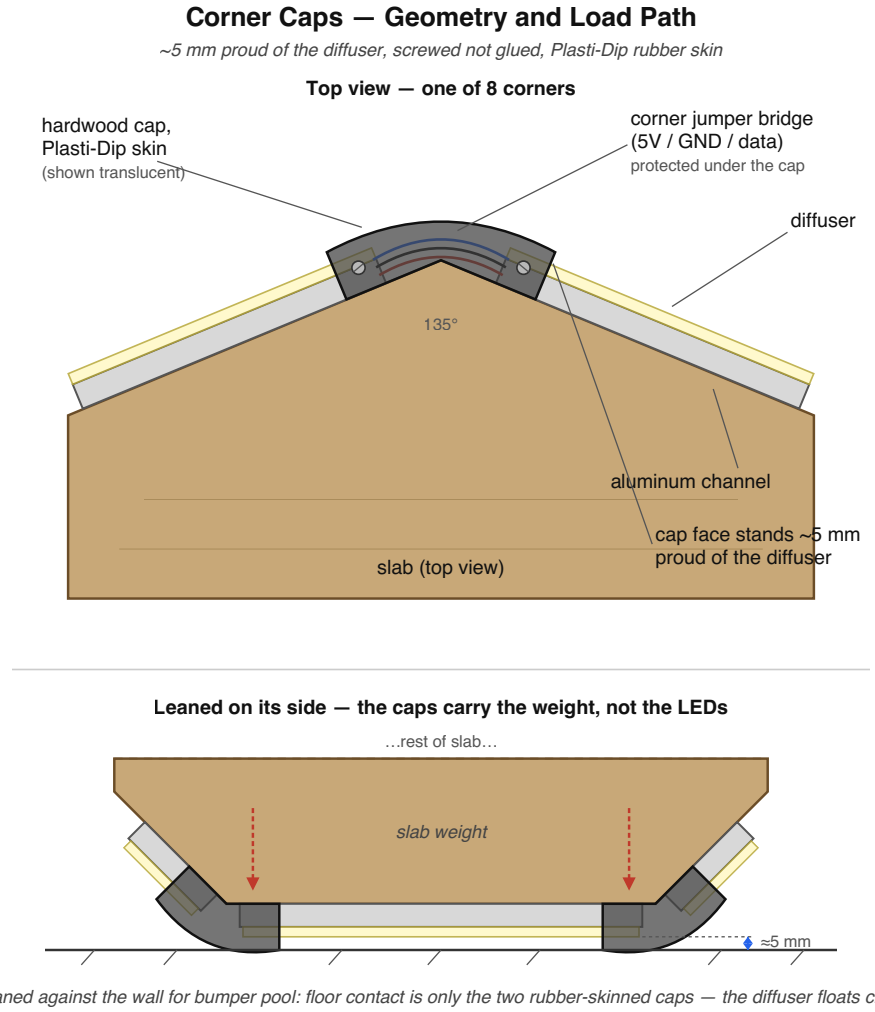



*Figure 4.2 — corner caps. Top: one of the 8 corners seen from above — the cap wraps the 135° corner, covers the jumper bridge in the channel miter gap, and stands ~5 mm proud of the diffuser. Bottom: the slab leaned on its side for bumper pool rests only on the two rubber-skinned caps; the channel and diffuser never touch the floor.*

Three ways to make them:

- **Plasti-Dipped hardwood caps (recommended)**: cut 8 blocks from scrap with the same 22.5° miter settings used for the channel, so each cap wraps its corner with two faces meeting at 135°. Ease all exposed edges with a 3/8" roundover, then dip or brush **3 coats of Plasti Dip** (dry ~30 min between coats, cure 4 h) before mounting — a soft, grippy, floor-safe rubber skin that adds impact absorption without changing the cap's geometry. Countersink two screws per cap into the slab edge, driven before dipping is easier: dip with the screws parked in their holes so the recesses stay open. Nearly free, matches the slab, fully covers the corner wiring.

- **Off-the-shelf**: wall-protection suppliers sell **135° corner guards** (Lexan or vinyl, sold as ~4 ft wall strips, ~$20–30) — one strip cut into 1"-tall pieces yields all 8 caps. These are thin profiles, so they cover the corner and the wiring but don't provide standoff; pair them with **7/8" screw-on rubber dome bumpers** (hardware-store item, ~$4 per 4-pack) — two per corner, one on each adjacent facet, supplying the actual proudness.
- **3D-printed**: custom 135° caps profiled around the exact channel cross-section, PETG (or TPU for built-in grip). Best fit if you have printer access.

**Making the hardwood caps without a router.** The cavity in the cap is not a blind pocket — the channel runs out both ends of the cap, so it's a **through cut**, which a circular saw handles. Two decisions make it easy:

- **Bias the channel toward one edge of the slab's outer face** rather than centering it. The cap's cavity then becomes a *rabbet* — two rip cuts on perpendicular faces and the waste falls out, nothing to chisel — and the leftover land is wide enough to take the mounting screws. A centered channel leaves ~3 mm of land on each side: no room for a screw, and the groove floor has to be cleared with stepped kerfs and a chisel between two walls.
- **Mill one long stick, cut the caps last.** Run the rabbet down ~1.2 m of scrap in a single setup, ease the show edges on that same stick with a block plane, and only then crosscut the 8 blanks. All the shaping happens on a long, clampable workpiece instead of sixteen small ones.

For the 135° corner itself, **kerf-and-fold beats a glued miter**: cut a 45° V-notch (180° – 135°) across the back of one blank, stopping 1–1.5 mm shy of the show face, glue the notch and fold it closed. The outer face is one continuous piece of wood with unbroken grain around the corner and no joint line, and it eases the corner as it bends. The only glue line is buried inside the corner, under three coats of Plasti Dip.

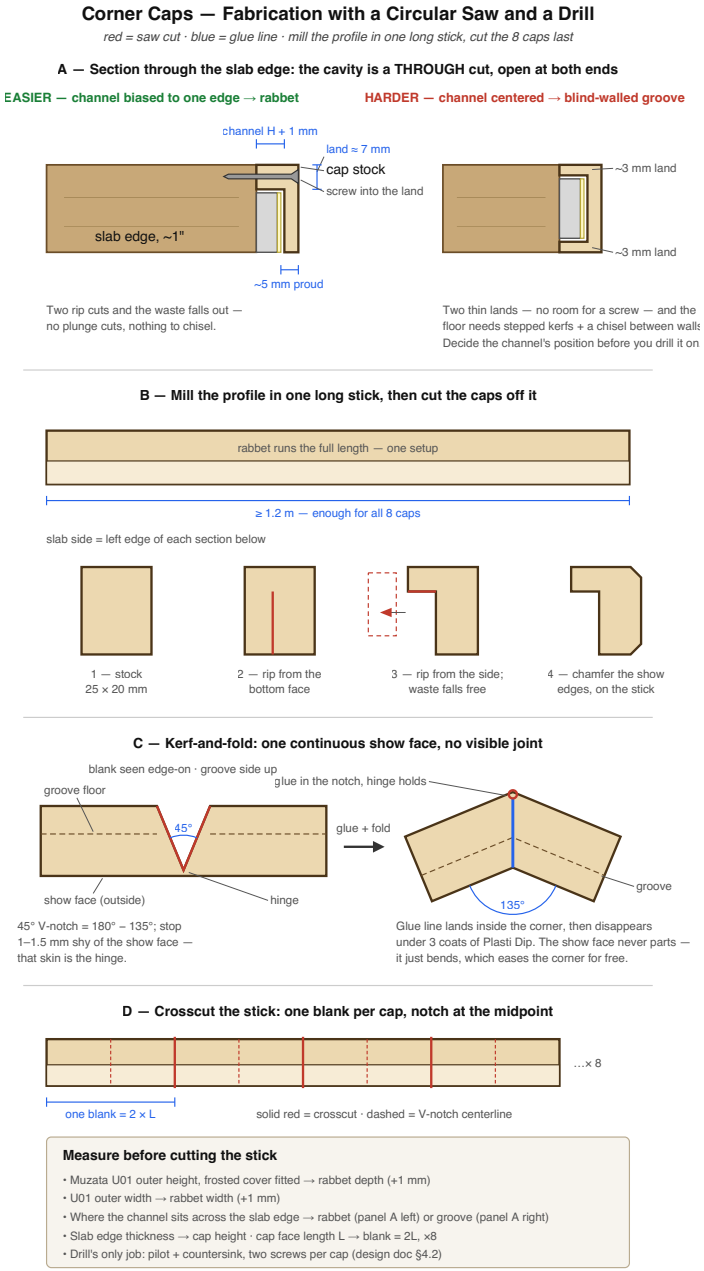



*Figure 4.3 — making the caps with a circular saw and a drill. A: the cavity is a through cut, so biasing the channel to one edge of the slab turns it into a rabbet (two rip cuts) instead of a walled groove. B: mill the profile down one long stick, chamfer it there too, and crosscut the caps last. C: instead of gluing two 22.5° miters, cut a 45° V-notch in the back of one blank and fold it — the show face stays continuous. D: crosscut layout and the dimensions to take off the real parts first.*

If the slab is thin enough to flex when leaned, add one more rubber bumper at the midpoint of each side, same proudness as the caps — six contact points per resting edge instead of two.

## 4.3 Piezo placement

One piezo per player's section (8 total), mounted near the **outer edge of its side** in a shallow bore drilled from the underside. Boring to within ~3–5 mm of the top surface leaves a thin "drumhead" of wood that flexes far more than the full slab thickness — a large sensitivity win — and puts the sensor directly under where players actually slap. The larger radius also spreads the sensors farther apart, which sharpens the firmware's biggest-jump-wins side discrimination. (Earlier revisions placed the discs at each wedge's centroid on the full-thickness slab; the rim bores replaced that after bench testing showed marginal sensitivity.)

Don't bore fully through to the laminate: a laminate-only floor is extremely sensitive but flexes enough to risk cracking the ceramic or de-bonding the disc over time. Avoid spots where the slab rests on the bumper pool frame; a clamped membrane can't vibrate.

**The input network lives at the disc, not the control board.** Each piezo gets its 1 MΩ bleed resistor and Zener clamp soldered right at the disc: twist the component legs and the pigtail wire into one bundle per node and flow solder once — two joints per piezo module. Zener band (cathode) toward the **signal** side; check before twisting. Make the joints in the pigtail ~1/2" off the disc (the ceramic hates long iron dwell), heat-shrink each node separately, then a dab of hot glue in the bore for strain relief. Clamping at the source means a slap's open-circuit spike dies at the disc instead of traveling the cable past seven neighboring runs. The signal wire then runs straight to the ADC pin — the optional 10–47 kΩ series-protection resistor is omitted in this build (the 1 MΩ + Zener are the real protection; the series R is extra hardening against a worst-case slam, easy to splice in later if wanted).

**Mounting**: bore floor clean and dust-free, cyanoacrylate or thin epoxy, brass face to the wood, even pressure for 30 seconds. Hot glue only for temporary tests.

**Wire routing**: one twisted pair (signal + GND) per piezo, 22 AWG stranded, routed in the underside kerfs to the control board — labeled S1–S8. Keep the pairs out of the kerfs carrying 5 V injection runs for their first few inches. Each pair carries an **inline JST-XH 2-pin a few inches off the control board** (same pattern as the strip harness): the disc end solders to the 1 MΩ/Zener network, the board end solders at the ADC hole + GND rail (A1–A8), and the JST midspan lets any disc — or the whole control box — unplug without desoldering. Label both halves of each connector S1–S8; the discs are identical, so a swapped plug just maps a disc to the wrong side (harmless, but confusing).

## 4.4 Cross-talk mitigation

Because the whole slab is one continuous piece of wood, every tap reaches all 8 piezos to some degree. The piezo directly under the tap reads strongest; piezos on adjacent sections read moderately; piezos on the opposite side read very weakly.

**First-line mitigation**: `readPiezos()` picks only the side with the *biggest jump above its own baseline* per scan as the accepted hit (the one exception is when the diametrically-opposite side also spikes in the same scan — that's the two-handed slap path). Adjacent cross-talk loses to the real hit automatically. Detection is

adaptive: each side's resting level is averaged at boot and tracked slowly, and a hit fires at baseline + `TAP_DELTA` — so per-side differences in piezo sensitivity and mounting are absorbed rather than fought with one global threshold.

**If you're getting false-side detection** (a tap on side 1 occasionally registers as side 2):

- Raise `TAP_DELTA` so weak cross-talk doesn't reach the fire level to begin with
- If that's not enough, edit `readPiezos()` to require the winning side's jump to be at least 2× the second-strongest side's jump before accepting the hit (you'll need to track the second-best delta alongside the best inside the scan loop)

**On the on/off gesture specifically**: this configuration is *especially good* for opposite-side detection. Cross-talk falls off with distance through the slab, and opposite sides are physically as far apart as possible. A single hard tap will never produce strong readings on opposite piezos — the slab simply doesn't transmit cross-talk that far at strong amplitudes. So "two strong readings exactly 4 sides apart" is unambiguously a deliberate two-handed slap.

## 4.5 Control box: enclosure options

Three choices, ranked from least to most effort:

**Option A — Project box, surface-mounted under the slab** (default). An ABS or aluminum project box (~112×62×31 mm — sized for the half-size control protoboard, see `doc-src/protoboard_half_layout.svg`) screws to the underside of the slab. Cable glands or grommets for entries. Cheapest, fastest, easiest to service. Slight visual penalty if you ever flip the slab over.

**Option B — Recessed pocket in the slab underside**. Route a rectangular pocket into the underside of the slab, sized to fit the protoboard + ESP32 + connectors. Cover plate (wood or thin metal) screws over it. Flush, invisible from below, but you have to commit to a location and dimension before routing — and any future hardware change means another router pass. Pay attention to slab thickness: with a ~1" slab and a typical protoboard depth of ~10–15 mm, you have very little wood left between the pocket bottom and the top surface. Best if you're confident in the design and the protoboard is thin enough.

**Option C — External enclosure on the bumper pool frame**. If you don't mind a small box mounted to the frame underneath rather than on the slab, you can keep the slab clean. Trade-off: now you have signal cables (8 piezo lines + 1 strip data line) crossing the disconnect along with power, which means a much larger multi-pin connector instead of the simple Powerpole DC pair. Not recommended for this build, but worth knowing it's an option if the slab needs to be perfectly minimal.

Whichever you pick:

- Leave room for **airflow** around any heat-generating components.
- Provide a **service access path** for the ESP32 USB port, even if you primarily flash via OTA. You may need wired serial for debugging.
- Make sure the **power switch** (on the frame, not the slab) is reachable without crouching all the way under the table.

## 4.6 Installation: removable top

The gaming table sits on top of an existing bumper pool table, and the top slab needs to be liftable. The build splits into two parts that connect via a single **DC quick-disconnect**. Mains AC stays put on the frame; only +5V DC crosses the joint.

> **This whole split is optional.** It exists to keep a mains-powered PSU on the frame. If you power the table from a USB powerbank on the slab (§3.5), there is no frame side and no disconnect — everything below is skipped, and the slab is just lifted off as one piece.

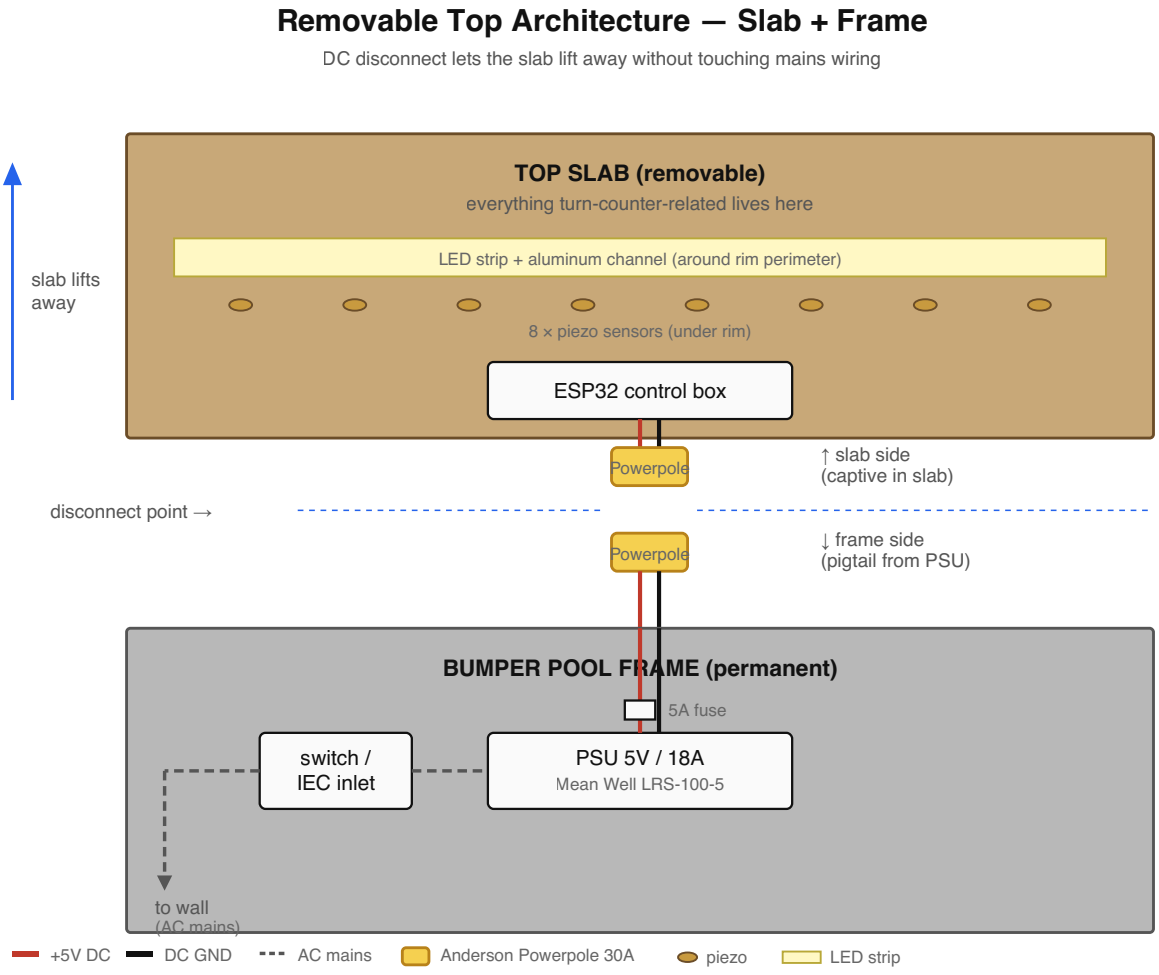



*Figure 4.6 — slab/frame architecture. Everything turn-counter-related lives on the removable top slab; the PSU and AC mains live permanently on the bumper pool frame. A single Anderson Powerpole DC connector is the only thing you disconnect when lifting the top.*

**Why DC and not AC at the disconnect**:

- *Safety*. AC mains at the disconnect point means a panel-mount connector that's potentially live every time you reach for it. Disconnecting mains at a casual touchpoint is a nope.
- *Practicality*. AC mains-rated panel connectors are bulky, expensive, and require strict polarity and earth bonding. Comparable DC connectors are cheap, robust, and idiot-proof.

**What lives where**:

*Slab (top, removable)*: LED strip + channel, all 8 piezos, ESP32 control box (with the optional level shifter if fitted), all signal wiring, and a DC pigtail terminating in a Powerpole connector.

*Frame (bottom, permanent)*: PSU, switch or IEC inlet, AC cord to wall, inline 5A fuse (DC side), DC pigtail terminating in the mating Powerpole connector.

**The disconnect itself**: **Anderson Powerpole 30A** connectors. They're polarized (can't connect backwards), genderless (the same housing mates with itself), rated comfortably above the 10A budget, and click together solidly. Red + black housing pairs are about $5; the contacts are soldered and heat-shrunk rather than crimped (decision locked in — `shopping_list.md` has the per-contact procedure), so there's no crimp tool to buy. XT60 connectors (RC vehicle world) are $2/pair and electrically equivalent, but the exposed bullet pins on the disconnected end aren't great if anyone ever pokes at them. Powerpole contacts are recessed.

**Cable**: 14 AWG silicone-insulated wire is overkill for 10A but stays flexible at the connector and is pleasant to work with. 16 AWG is the practical minimum. Run twisted pairs (red 5V + black GND) for tidy routing. Length: PSU location to slab + 18" of slack so you can lift the slab a foot before having to disconnect.

**Slab cable entry**: Buy the rubber grommet first, *then* drill the hole to match — most hardware-store grommets are spec'd by their *panel-hole size* (commonly ½″ or ⅝″), and the grommet's inside diameter should snug-fit your DC cable (the 14 AWG silicone twisted pair is ~$\frac{5}{16}$″ / 8 mm OD). Use a soft rubber grommet from the hardware store, not a sheet-metal-panel bushing like the Heyco 2092 — those snap-fit into 3 mm panels and won't retain in 1″ wood. After installing the grommet, secure the cable inside the slab with a P-clip or adhesive cable mount within 4 inches of the entry — any pull on the cord should hit the clip, not the connector terminations.

**PSU mounting**: The Mean Well's mounting flanges screw to a small wood block (~4×6×1"), which then screws to the bumper pool frame underside in a ventilated location. Don't seal it inside an enclosed box — these run warm and need airflow.

**Bench testing**: When the slab is off the table, the lights have no power. Keep a spare Powerpole pigtail wired to a 5V/3A bench supply — plug into the slab's connector for moderate-brightness operation away from the table. Not enough current to drive the strip hard, but plenty for testing.

# 5. Phase-by-Phase Build Steps

## Phase 1: Plan & Gather

☐ Phase 0 complete and Tap Light works (see §0)

☐ Confirm BOM and order remaining parts

☐ Print this doc and the wiring diagram (`turn_counter_wiring.svg`), keep them on the bench

☐ Measure the actual slab: confirm it's ~1" thick, all 8 sides equal length, lies flat on the bumper pool frame

☐ Identify a ventilated mounting location on the underside of the bumper pool frame for the PSU (and note where the AC outlet is relative to it)

☐ Plan the cable run from PSU → Powerpole → up to the slab. Mark roughly where the slab cable will enter

☐ Confirm you have wall outlet access within reach of the planned PSU location, ideally on a switchable surge protector

## Phase 2: Bench Prototype

☐ Wire ESP32-S3 + 30-LED test segment + 1 piezo on breadboard per the wiring diagram (`doc-src/breadboard_layout.svg` shows a hole-by-hole placement). Connect the piezo to GPIO 1 (`PIEZO_PINS[0]`)

☐ Use the purpose-built bench sketches rather than editing the main firmware — there is one for each thing you want to prove, and none of them needs a temporary edit you have to remember to undo:

- `make flash-tap` (`tap_light`) — strip smoke test and the per-side LED calibration. On a short bench strip this is the one to use; it doesn't assume a full octagon.
- `make flash-piezo` (`piezo_test`) — per-channel tap diagnostics. With a single piezo wired, tap it and watch which channel reports.
- `make flash-eight` (`eight`) — all eight seats, clockwise passing, no setup mode. The simplest end-to-end proof once the real table is wired.

☐ With one seat's piezo wired and `eight` flashed, tapping it advances the turn to side 1 exactly once; further taps on that same piezo print *"Tap on side N ignored - not the current seat"*, because the turn has moved on. That message is the current-seat filter doing its job, not a fault

☐ If a tap lights the wrong seat, do **not** edit `PIEZO_PINS` — run `make map-piezos`, tap each side as it lights, and the corrected map is stored on the board

☐ Open Serial Monitor at 115200 baud

☐ Add a temporary `int v = analogRead(PIEZO_PINS[0]); if (v > 100) Serial.println(v);` inside `loop()` and tap the piezo — the threshold of 100 keeps the serial output quiet at idle and only prints when something interesting happens (note the single-read pattern: a piezo's voltage decays fast, so reading twice would give two different numbers)

☐ Note the reading at rest (probably 0–50) and the peak when tapped (likely 500–3000+)

☐ Set `TAP_DELTA` to ~30% of the peak reading's jump above the printed baseline — adjust by feel (the boot log prints each side's seeded baseline; a tap fires at baseline + `TAP_DELTA`)

☐ Confirm the LEDs change color/zone on tap before proceeding

☐ Test power-cycle persistence: tap a few times, unplug, replug, verify state restored

☐ If using OTA: configure Wi-Fi credentials in firmware, confirm device appears on the network and can be reached via `ping turn-counter.local`

**Don't move forward until this works reliably.**

## Phase 3: LED Rim

☐ Measure all 8 sides — confirm they're actually 20", not 19.875" or something

☐ Cut aluminum channel: 8 pieces with 22.5° miters at each end

☐ Pre-fit the channel dry around the slab's outer edge, mark mounting screw locations

☐ Cut LED strip into 8 segments, ~30 LEDs each, between pad gaps

☐ Lay strips into channels, peel adhesive backing

☐ Solder jumper wires at each corner (3-conductor: 5V, GND, Data); heat-shrink each joint

☐ Test continuity: data line from start to end of the assembled strip with multimeter

☐ Mount channel pieces to the **outer edge of the slab** with screws (or recessed in a routed rabbet — see §4.1)

☐ **Calibrate the per-side LED counts.** The firmware no longer assumes a uniform `LEDS_PER_SIDE` — each side's actual count lives in a `sideLedCounts[8]` table (corner cuts make sides unequal). Flash `tap_light`, open the serial monitor, and use the calibration commands (`0`–`7` select a side, `+`/`-` move its boundary, `p` prints the table) until every color flip in the side-colors mode sits exactly on a physical corner. The table persists in NVS and `turn_counter` reads it automatically; also paste the printed table into both sketches' `sideLedCounts` defaults so it survives a flash-erase

☐ Solder pigtails for power injection at three points along the strip: at the start (DIN end), at the corner roughly halfway around (the corner between the 4th and 5th side as the strip runs), and at the end of the strip. Each pigtail joins +5V and GND from the slab's main DC rails to the strip's 5V and GND pads

☐ Mount the 8 corner caps (§4.2) over the corner jumper bridges — screwed, rubber-faced, ~5 mm proud of the diffuser

## Phase 4: Piezo Mounting

☐ Drill 8 rim bores from the underside, one per side near the outer edge, leaving a ~3–5 mm floor (§4.3) — verify remaining thickness before the first glue-down, and never bore through to the laminate

☐ Solder leads to all 8 piezos (red to +, black to –). Quick — under 2 seconds per pad

☐ Build each disc's network in the pigtail ~1/2" off the disc: twist the 1 MΩ + Zener legs into the wire bundle, one soldered joint per node, Zener band toward signal; heat-shrink each node separately (§4.3). Terminate the disc pigtail in its inline JST-XH 2-pin, labeled S1–S8

☐ Test each piezo before mounting: connect to multimeter on AC voltage, tap it, expect a brief swing

☐ Glue each piezo into its bore, brass face against the wood floor — vacuum the drilling dust out first

☐ Route each twisted pair through the underside kerfs toward the control box; secure every 6"; label S1–S8

## Phase 5: Control Box

☐ Build the protoboard: half-size Perma-Proto — ESP32 socket (use female headers, don't solder the module), 1000 $\mu$F cap, and the GPIO 11 → 470 Ω → strip data run (470 Ω spliced in-line, near DIN). Direct 3.3 V drive — **no level shifter** by default (cols 23–30 stay empty; that's where the optional '125 drops in, §3.3). Print `doc-src/protoboard_half_layout.svg` for placement; `doc-src/protoboard_wiring.svg` is the retired full-size map, kept only for the DevKitC-1 v1.1 pin reference. Piezo networks are **not** on this board — they live at the discs (§4.3)

☐ Solder the 8 labeled piezo pairs into their row-a/j holes per the layout — signal to the GPIO column, GND to the nearest rail. No series resistor, no board-mounted JST headers; instead each pair gets an **inline JST-XH 2-pin a few inches off-board** for service (same pattern as the strip). Label the connectors S1–S8

☐ Solder the 3-wire strip harness (5V / data / ground) directly to the board, with an in-line JST-XH 3-pin a few inches off-board as the service disconnect (the board feeds the strip-start injection point; the mid and end taps stay on the slab rail)

☐ Add a 5V/GND input pigtail (its far end lands in a WAGO 221 lever node on the slab DC rail)

☐ Test before installing: power up with the bench-test pigtail (see §4.6) connected to a 5V supply — either a small bench supply, or a USB-C wall adapter with a USB-C-to-Powerpole adapter cable, or even just the ESP32-S3's USB power if you only need to verify the firmware logic without driving the LED strip — then run firmware and tap piezo leads with a screwdriver to fake hits

☐ Mount protoboard inside the chosen enclosure (see §4.5 — typically a project box under the slab); cut cable entries for power and signal

## Phase 6: Final Assembly

The build splits into frame-side and slab-side work — see §4.6 for the architecture.

### Phase 6a — Frame side (permanent installation)

☐ Mount Mean Well PSU to the wood block with #8 screws through its mounting flanges

☐ Screw the wood block to the underside of the bumper pool frame, in a ventilated location

☐ Wire AC mains using the discrete-component approach (BOM default): wall plug → IEC C14 inlet → SPST rocker switch (interrupts Line only) → PSU's L (live) terminal. No fuse on the AC line — the inline blade fuse is DC-side only (see §2's wiring note; the PSU has its own internal input fuse). Neutral runs IEC inlet → PSU's N terminal **direct, with no switch** (the SPST switch has only one pole — it breaks Line only, same as every wall switch in a US household). Earth runs IEC inlet's Earth pin → PSU's chassis Earth lug, no switch on Earth. The IEC inlet and switch terminals are 0.250″ faston tabs — either crimp female spade connectors onto the wire ends or solder directly with heat-shrink (3 on Line + 1 Neutral + 1 Earth = 5 faston connections total). Verify continuity with a multimeter (switch ON: Line continuous from inlet to PSU; switch OFF: Line open; Neutral and Earth always continuous) **before** plugging into the wall. Remember the safety habit from §2: the SPST switch breaks Line only, so **unplug from the wall before opening the enclosure** — only the wall plug guarantees both Line and Neutral are dead. If you'd rather interrupt both poles, swap the SPST for a DPST switch (wire Neutral through its second pole) or use a Schurter DG12 integrated inlet — see §2 wiring note

☐ Wire PSU's +V output through the inline 5 A fuse to the +V contact of a Powerpole housing; PSU's –V direct to the –V contact

☐ Solder Powerpole contacts onto silicone wire ends (procedure in `shopping_list.md`); insert into housings (red = +V, black = GND); slide the two housings together so they're locked

☐ Leave the cable long enough to reach the slab's connector with ~18" of slack

☐ First power test (frame only, no slab connected): switch on, multimeter the Powerpole, expect +5V between red and black

☐ Switch off before continuing

**Phase 6b — Slab side (mobile assembly)**

☐ Drill a 1/2" hole near the slab edge for the cable; install the rubber grommet

☐ Run the slab-side DC cable through the grommet into the slab

☐ Solder Powerpole contacts onto the slab-side cable ends; insert into mating housings (the genderless design means it's the same housing as the frame side)

☐ Inside the slab: branch from the slab-side DC cable to the control box's 5V/GND terminals AND to the 3 strip injection points (start, middle, end of strip). Branching method: land all incoming wires in the BOM's WAGO 221 lever connectors (cleanest), or twist-and-solder them with heat-shrink (works fine but harder to service)

☐ **Verify polarity at every junction with a multimeter before connecting.** Reversed 5V instantly kills the strip

☐ Secure the cable with a P-clip or adhesive cable mount within 4" of the grommet — strain relief should hit the clip, not the connector

☐ Connect all 8 piezo JST connectors to the control box

☐ Connect the strip data + ground JST

☐ Mount the control box; dress all wiring with cable clips; nothing dangling

**Phase 6c — Mate and test**

☐ **Slab bench test (before connecting to the frame)**: with the slab on a workbench (sawhorses or table), connect the bench-test Powerpole pigtail to the slab's connector and to a 5V supply. Confirm the strip lights up and the firmware advances on tap. If anything's wrong, fix it now while access is easy

☐ Disconnect the bench supply

☐ Mate the slab Powerpole to the frame Powerpole

☐ First full power-up: switch on, watch for smoke (good sign: no smoke)

☐ Tap through every player position in turn (with the default 4-player layout, that's 4 advances; each player owns 2 contiguous sides). Look for dead LEDs, wrong colors, or visible gaps at corner joints — the lit zone should sweep cleanly across all 8 sides as you cycle through

☐ Test the disconnect: switch off → unmate Powerpole → lift the slab a foot → set it back → reconnect Powerpole → switch on → confirm everything works

## Phase 7: Tune & Test

☐ Re-flash with `Serial.println` instrumentation enabled (over OTA if Wi-Fi is set up, or USB). For calibration, temporarily add a print of the raw `analogRead` values for *every* side that exceeds a low threshold (say, 100) — this captures both direct hits and cross-talk so you can see the cross-talk geometry

☐ Open the Serial Monitor (or for OTA, a network log viewer) and keep it visible while you tap

☐ **For the per-side reading test**: just tap each side firmly while watching the Serial Monitor. The instrumentation prints raw readings independent of the active-side filter, so you don't need to disable it — peaks for all 8 sides should be roughly similar

☐ If any side reads much lower, check the glue contact and lead solder joints

☐ Set `TAP_DELTA` to the value that ignores incidental table bumps but catches deliberate taps (typically ~30% of a deliberate tap's jump above baseline)

☐ **Cross-talk test**: tap side 1 hard, look at the serial output. The strongest reading should be side 1; sides 2 and 8 (adjacent) will show smaller readings; side 5 (opposite) should be near-zero. Confirm the firmware identifies the hit as side 1 — the only positive signal is the lit-zone advance (or the absence of an "ignored" log line for that side). If the firmware ever picks a non-adjacent side as the strongest, raise `TAP_DELTA` or add a relative-strength check (see §4.4)

☐ **On/off gesture test**: with two hands, slap two opposite sides simultaneously. Lights should toggle. If no toggle, check that both piezos spike above their baselines; raise `OPPOSITE_PAIR_WINDOW_MS` if your slap timing isn't quite synchronized

☐ **Setup gesture test**: rapidly tap an **unlit** side 4 times within 2 seconds. Strip should start blinking. Tap once more to cycle player count. Wait 3 seconds; strip resumes normal play at player 1

☐ Set `BRIGHTNESS` to your preferred level (start at 128, adjust)

☐ Play an actual game. Take notes on anything weird

# 6. Controls & Setup Mode

Once installed, the user-facing controls are:

| Action | Result |
|---|---|
| Tap your **own** (lit) side during play | Advance turn to next player |
| Tap a side that *isn't* lit | Nothing — only the active player can pass turn |
| Rapid-tap an **unlit** side 4 times within 2 seconds | Enter setup mode (sides flash) |
| Tap two **opposite** sides simultaneously (a two-handed slap, on sides directly across from each other) | Toggle on/off |
| In setup: tap any side | Increment player count by one each tap, wrapping 8 → 2 (starts from whatever count was previously saved) |
| In setup: stop tapping for 3 s | Save player count, exit setup, reset to player 1 |
| Power cycle | Resume on the same player and same on/off state (state persists) |

Number of *blinking* sides = current player count selection.

**The active-side rule**: turn advance only triggers when the tapped side belongs to the currently lit zone. If player 3 is active, only taps within player 3's wedge advance the turn. This means players can't accidentally (or deliberately) advance someone else's turn by tapping their own section while waiting. It also reinforces "look at the lights to know whose turn it is" as the natural way to play. Non-active-side taps are silently ignored — no flash, no sound, the lit zone stays put.

This rule applies only to turn advance. The on/off gesture works from any sides (any opposite pair). Setup-mode entry, by contrast, counts **only taps on unlit sides** — so fast turn-passing (which always lands on the lit side) can never trip it. See below.

**Why the entry gesture isn't a "hold"**: piezos detect vibration, not pressure. They produce a brief voltage spike when struck and then return to baseline — there's no signal to read while you keep your hand there. So setup entry uses a rapid tap burst instead. To keep a brisk round of normal play from looking like that burst, the firmware counts **only taps on unlit (non-active) sides** toward the gesture — turn-passes always land on the lit side, so they never accumulate. The natural way in is therefore to rapidly tap any dark side 4 times. If you instead start on your own lit section, the first tap just passes the turn (your side goes dark), and four more dark-side taps then trigger setup — i.e. it takes one extra tap from your own seat. Exiting setup resets to player 1 anyway, so that initial pass doesn't matter.

**The on/off gesture in detail**: any two-handed slap on diametrically opposite sides toggles the LEDs. The firmware buffers each tap for 150 ms before committing it as a turn advance — long enough to detect a near-simultaneous opposite tap, short enough that the resulting delay on normal turn passes is barely perceptible. Off state is fully dark; the device still polls the piezos and watches for the on-gesture, but ignores everything else. The 150 ms buffer also handles the asymmetric-strike case where one hand lands a few milliseconds before the other.

**Why opposite sides specifically**: cross-talk through the wooden slab falls off with distance. Adjacent piezos (sides 1 + 2) might both register from a single hard hit on side 1. Opposite piezos (sides 1 + 5) physically can't — they're as far apart as possible in the slab, so two strong simultaneous readings on opposites can only come from a deliberate two-handed slap.

# 7. Firmware

The main firmware is in `turn_counter.ino`. The Phase 0 starter firmware is in `tap_light.ino`. Key tunables at the top of the main file:

| Constant | Default | Tune to… |
|---|---|---|
| `NUM_LEDS` | 240 | Buffer ceiling only. The lit total is `totalLeds()`, the sum of the calibrated per-side table |
| `sideLedCounts[8]` | calibrated | Per-side LED counts, set by `tap_light` 's serial calibration and stored in NVS `"octagon"/"sides"` . Replaced the old uniform `LEDS_PER_SIDE` , since corner cuts make sides unequal |
| `sidePiezoPin[8]` | `PIEZO_PINS` order | Which ADC pin serves each side, stored in NVS `"octagon"/"pmap"` . Set by `make map-piezos` , not by editing source |
| `BRIGHTNESS_DEFAULT_PCT` | 50 | Startup brightness before any phone change; the runtime value lives in NVS `"octagon"/"bri"` |
| `BRIGHTNESS_MIN_PCT` | 5 | Slider floor. Going fully dark is the on/off control's job, not the slider's |
| `BRIGHTNESS_SETTLE_MS` | 2000 | Quiet time before a brightness change is written to flash, so dragging the slider costs one write, not twenty |
| `BRIGHTNESS_FRAME_MS` | 16 | Fade step interval (~60 Hz). A `show()` blocks ~7 ms for 221 pixels, so this can't run every loop pass |
| `BRIGHTNESS_FADE_ALPHA` | 40 | Lerp rate out of 255 per frame. Settle time scales with distance: ~130 ms for a 1% nudge, ~540 ms for a full 5→100% sweep |
| `TAP_DELTA` | 720 | Adaptive: a tap fires at a side's auto-tracked baseline + this delta. Set from the 2026-07-28 bench recordings in `data/piezo/` — top of the gap between the soft-tap cluster (<490) and the weakest real tap (757), with ~12× headroom over the worst idle noise seen (59) |
| `DEBOUNCE_MS` | 250 | Raise if double-triggers, lower if it feels sluggish |
| `SETUP_TAP_COUNT` | 4 | Taps on one side to open setup — or, while the table is off, to wake it |
| `SETUP_TAP_WINDOW_MS` | 2000 | Window in which those taps must occur |
| `MODE_DEMO_MS` | 5000 | Setup phase 1: how long each mode demos before the dial advances |
| `MODE_ABORT_IDLE_MS` | 25000 | Setup phase 1: no tap for a full demo rotation aborts, changing nothing |

| Constant | Default | Tune to… |
|---|---|---|
| `SETUP_JOIN_IDLE_MS` | 5000 | Setup phase 2: idle time before the roster is committed and setup exits |
| `OPPOSITE_PAIR_WINDOW_MS` | 150 | Window for detecting the on/off two-handed gesture; raise to make it more forgiving, lower for snappier turn passes |
| `WIFI_SSID` / `WIFI_PASSWORD` | (unset) | Your home Wi-Fi, in `firmware/turn_counter/secrets.h` (gitignored; copy `secrets.example.h`). Unset = radio off |
| `OTA_HOSTNAME` | `turn-counter` | In `secrets.h`; mDNS name, reach device at `turn-counter.local` |
| `OTA_PASSWORD` | `change-me` | In `secrets.h`; required to push firmware updates |
| `WIFI_RETRY_MS` | 30000 | How often a down link is retried; OTA starts whenever the link comes up |

`PLAYER_COLORS[]` array defines the color for each player — edit to taste.

**Build environment**: Arduino IDE with ESP32 board support, FastLED library installed (ArduinoOTA is bundled with the ESP32 core). Board: **"ESP32S3 Dev Module"**. **USB CDC On Boot: Enabled** (so Serial works over the native USB port). Upload speed: 921600.

## 7.1 OTA firmware updates

The device tries to join Wi-Fi at boot with a 5-second timeout. If it joins, OTA is available; if not, it just runs locally — the device works fine without a network.

**To push an update from Arduino IDE**:

1. With the device powered and on the same network as your computer, the IDE's "Port" menu will list `turn-counter at 192.168.x.x` (or whatever the mDNS resolves to) under "Network Ports".
2. Select that port, hit Upload, enter the OTA password when prompted.
3. The strip goes dark, then a blue progress bar fills around the rim as the update transfers.
4. On success, all LEDs flash green briefly. On failure, all red for 2 seconds.
5. Device reboots into the new firmware.

**To push an update from the command line**:

```
make ota                    # compiles, resolves turn-counter.local, checks it's reachable, pu
make ota HOST=192.168.1.42  # same, when mDNS won't resolve
```

`make ota` reads the OTA password out of `secrets.h`, so it never lands in a Makefile or in shell history. It resolves the hostname and confirms the board answers a ping before it starts transferring — the difference between a clear "can't reach it" and a silent 60-second hang.

Note that OTA's port 3232 is **UDP**, not TCP: espota sends an invitation datagram and the device connects back over TCP to receive the image. UDP has no handshake, so there is nothing a `connect()` could confirm — a TCP probe of 3232 fails against a perfectly healthy board. The pre-flight therefore checks reachability, and leaves "is OTA actually listening" to espota's own response (and to the `OTA ready` line in the boot output).

The device also retries a failed or dropped Wi-Fi connection every 30 seconds and starts OTA the moment it joins, so a router that was slow or down at boot no longer means power-cycling the table to get OTA back.

**Security note**: anyone on your local network can attempt to flash the device. The OTA password protects against accidental or casual attacks but isn't strong security. Don't put this on a network with untrusted devices, and definitely don't expose it to the internet.

**If OTA breaks** (most common cause: pushing firmware that crashes immediately and loses Wi-Fi): you can always fall back to USB. Plug into the ESP32-S3, hit Upload, done. Keep a USB cable accessible.

## 7.2 Phone control

The board serves a single self-contained page on port 80 — no app to install, and no external assets (it has no internet path, so anything external wouldn't load anyway).

| Method | Path | Purpose |
|---|---|---|
| GET | `/` | the page |
| GET | `/api/config` | mode names, seat colours, side count — fetched once |
| GET | `/api/state` | mode, brightness, lit, current seat, roster, ready, setup — polled every 2 s |
| POST | `/api/mode?value=0..3` | set mode; 409 while a tap-setup session is running |
| POST | `/api/brightness?value=5..100` | set brightness |
| POST | `/api/power?value=on` or `off` | light or darken the table |

Range errors are 400; 409 means the request was valid but the table is mid-setup, and the person physically at it wins.

A mode change from the phone leaves the roster alone — it isn't a setup session. Off preserves all game state and isn't persisted, so the table always boots lit; four fast taps on one side wake it without a phone.

Brightness eases rather than snapping. The strip lerps toward the target the way a web animation would — `current += (target - current) * alpha` each 16 ms frame — instead of running a fixed-duration tween, because the slider retargets up to four times a second while dragging and a tween would restart, and visibly stutter, on every one of those. Smoothing simply follows, and eases out for free.

Two details make it work. `FastLED.setBrightness()` does nothing until the next `show()`, and in steady play nothing redraws, so the fade tick issues its own `show()` — re-sending the frame already in `leds[]` at the new scale, with no re-render. And it lands the moment the residual drops below one raw unit, since the strip can't display finer than that; without that snap a 1% nudge kept firing `show()` for ~600 ms after it was visually finished.

The flash write is separately deferred two seconds past the last change, so dragging the slider costs one NVS write rather than one per step.

```
curl -s http://turn-counter.local/api/state
curl -s -X POST "http://turn-counter.local/api/power?value=off"
```

**Reaching it**: `turn-counter.local` works on Apple devices. Android has no system-wide mDNS resolver, so use the IP — give the board a DHCP reservation (its MAC is printed at boot) and `make qr URL=<ip>` prints a sticker for under the table.

**Security**: no authentication, same posture as OTA. Anyone on the LAN can change mode, brightness and on/off. Nothing destructive is exposed — no OTA trigger, no NVS wipe, no piezo remap.

# 8. Troubleshooting

| Symptom | Likely cause | Fix |
|---|---|---|
| LEDs flicker or first LED is wrong color | Weak data signal into pixel 1 (3.3 V direct drive is marginal) | Confirm 470 Ω + common ground; if it persists, add the optional 74AHCT125 level shifter (§3.3) |
| Far-end LEDs tint pink/orange | Voltage drop along strip | Add or check power injection at midpoint and end |
| Whole strip dark | PSU off, switch off, blown fuse, or reversed polarity | Check switch first, then fuse, then PSU output, then polarity |
| Tap doesn't register | `TAP_DELTA` too high, bad piezo solder, glue not contacting wood | Lower `TAP_DELTA`, reflow joint, re-glue |
| Tap on side 1 lights side 3 | Piezo wire mapping wrong | Run `make map-piezos` — each side lights in turn, you tap it, the corrected map is stored on the board. No reflash, no wire tracing |
| Adjacent sides cross-trigger | Mechanical cross-talk through wood | Foam break, kerf cut, or relative-strength filter in firmware |
| Weird boot behavior | Strapping pin pulled wrong | Confirm GPIO 0/3/45/46 unused (ESP32-S3 strap pins) |
| ESP32-S3 won't flash via USB | Upload speed, USB CDC setting, or held button | Drop to 115200 baud, confirm Tools → USB CDC On Boot is Enabled, hold BOOT + tap RESET + release BOOT to enter download mode manually |
| OTA port doesn't appear in IDE | Wi-Fi didn't join, wrong network, mDNS not resolving | Check Serial Monitor at boot for IP; ping `turn-counter.local`; fall back to USB |
| Phone page loads on iPhone but not Android | Android has no system-wide mDNS resolver, so `.local` never resolves | Use the IP. DHCP-reserve it on the router (MAC is printed at boot) so it stops moving, then `make qr URL=<ip>` |
| Table dark and taps do nothing | Someone turned it off from a phone | Four fast taps on one side wake it; or power-cycle, since off is never persisted |
| OTA fails midway | Network drop or insufficient flash | Retry; consider partition scheme with larger OTA region |
| Player count resets unexpectedly | NVS partition issue | Erase flash and re-flash: `esptool.py erase_flash` |
| Solder joint won't take | Tip too cold, no flux, dirty surface | Increase iron temp, add flux pen, clean the surface with isopropyl |
| Solder joint looks dull / blobby | Cold joint — moved while cooling, or insufficient heat | Reflow with flux, hold steady ~2 sec while cooling |

# 9. Future Enhancements

If the basic build lands well, things worth considering later:

- **Turn timer**: each player has N seconds; the lit zone slowly drains around the side as time runs out. Tap before it empties or it auto-advances with a red flash.
- **Companion app**: ESP32-S3 already has Wi-Fi up for OTA. Run a small web server, expose a phone-friendly page for round counting, scoring, initiative tracking. Especially useful for RPGs. (BLE 5.0 is also on the table — the S3 has both radios, and ADC1 is unaffected by either.)
- **Audio feedback**: a small piezo *buzzer* (different from the sensor piezos) on a free GPIO for soft turn-pass clicks.
- **Per-game profiles**: store named configurations in NVS. "Magic 4-player", "RPG initiative for 6", etc.
- **Initiative mode**: instead of round-robin, player order is a stored sequence. Useful for D&D-style turn order that isn't seat-based.
- **Color customization**: the companion app could let players pick their own colors before the game starts.
- **Round counter**: subtle flash or pulse when a full round completes. Easy to add given current architecture.

# 10. Quick Reference Card

Keep this part handy at the table:

> **To pass turn**: tap the lit section in front of you. Only the active player can advance. **To turn on/off**: slap any two opposite sides of the table at the same time (a two-handed gesture: left hand on one side, right hand on the side directly across). **To change player count**: rapid-tap an unlit (dark) side 4 times within 2 seconds. The lights will start blinking. Then tap to cycle through counts. Stop tapping for 3 seconds to save. **To reset to player 1**: enter setup mode and exit (count stays the same, turn resets). **State persists across power cycles** — including on/off, current player, and player count.

---

*End of document.*